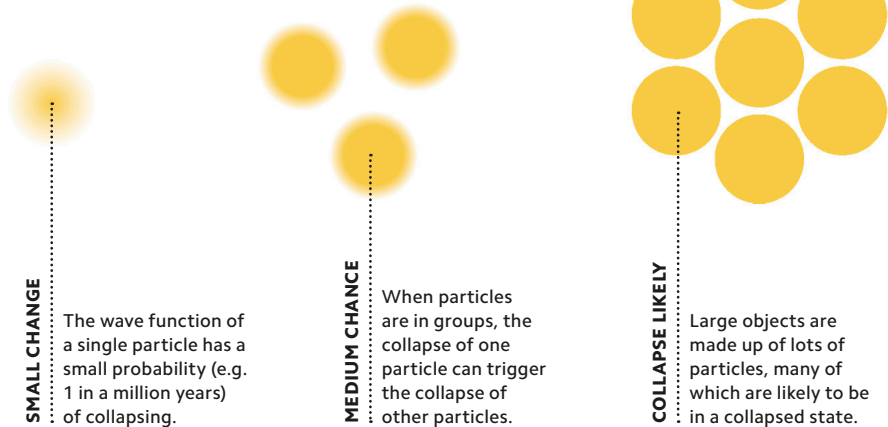


Wave function half-life

Though initially uncertain, a wave function has a certain probability of collapsing spontaneously, somewhat like the half-life of radioactive decay. Collapse may be accelerated by interactions with the collapsing wave functions of other particles.



SPONTANEOUS COLLAPSE

The Copenhagen interpretation's suggestion that measurement or observation trigger the collapse of the wave function raises the troubling question: what happens when no measurements are taken? Alternative or spontaneous collapse theories get around this by suggesting that the collapse happens automatically and randomly, without a trigger from the outside world. A wave function's likelihood of collapse can still be influenced by the wave functions of particles surrounding it.